

Santiago Avilez Ariza

Senior Full-Stack Software Engineer | Production AI features · TypeScript · Node.js · NestJS · React · Next.js
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Senior Full-Stack Engineer with 5+ years owning production systems end-to-end in TypeScript, React, Next.js and NestJS. I work from problems, not tickets: I define the architecture, make the trade-offs, ship, and measure the outcome. Built public-facing platforms serving 100,000+ users, shipped production AI features (LLM tool calling, RAG, MCP), designed multi-tenant systems with Postgres row-level security, and mentored less experienced developers along the way.

EXPERIENCE

Full-Stack Engineer — City of Neuquén - Digital Modernization Office, Neuquén, Argentina, Feb 2025 – Present

- Own architecture and technical direction for public-facing platforms (React + NestJS + PostgreSQL) serving 100,000+ users. I define system design, evaluate trade-offs, and drive delivery from problem definition to production with full autonomy.
- Mentor junior developers through code reviews, pairing, and architecture guidance, raising code quality across the team.
- Led frontend performance refactor — code-splitting, lazy loading, rendering optimizations — reducing load times by **30%** and boosting form completion by **15%**
- Adopted AI-first development workflow using Claude Code and Cursor — spec-driven development (SDD: design, task and spec docs) with orchestrator/sub-agent patterns, reducing implementation cycle time on complex features.
- Built an MCP server (TypeScript MCP SDK, stdio transport) that indexes the organization's repositories (code, commits, PRs and docs) for RAG, letting developers query the codebase from AI assistants. github.com/santiagoavilez/mcp-repos-rag
- Established automated testing with Jest (~85% coverage on critical modules), significantly reducing regressions across production releases.

Stack: TypeScript · React · Next.js · NestJS · Redis · Node.js · PostgreSQL · TypeORM · Docker · Jest · REST APIs

Full-Stack Engineer — Zoada Dev Studio (remote) Dec 2020 – Jan 2025

- Built and shipped full-stack applications (React + Next.js + NestJS) for clients in media, e-commerce, education, and startups, owning end-to-end delivery from DB modeling to production deployment.
- Reduced server-side response times by 40% through NestJS query optimization and API response restructuring on a client platform.
- Chose Vertical Slice Architecture over the studio's default Clean Architecture for an e-commerce platform serving 100,000+ users, organizing code by feature to reduce cross-layer coupling and speed up feature delivery.
- Integrated payment gateways and third-party APIs, building RESTful services with error handling, retries, and logging under real production load.

Stack: TypeScript · JavaScript · React · Next.js · NestJS · Node.js · MySQL · PostgreSQL · Jest · REST APIs · Clean Architecture

PROJECTS

Fulbbo — Lead Engineer | fulbbo.com

- Designed a geographic cache system over Google Places API (Supabase PostgreSQL + 5km grid), reducing API costs by ~90% in high-traffic zones.
- Implemented row-level security (RLS) policies in Supabase Postgres to enforce per-venue data isolation across all API operations—ensuring zero cross-tenant data leakage without application-layer filtering logic.
- MercadoPago OAuth payments per venue, with idempotent webhook handling.
- Built an AI search layer (OpenAI GPT-4o mini, function calling, Zod) turning natural-language queries into validated filters, cached in Redis.
- Built a WhatsApp booking chatbot (Anthropic SDK + OpenAI; Claude Haiku / GPT-4o mini chosen per task by cost and latency, tool calling, Zod) mapping free text to allow-listed actions.

Stack: TypeScript · Next.js · Node.js · tRPC · Supabase (Postgres + Realtime) · Redis · OpenAI · Zod · Vercel · Vitest

TECHNICAL SKILLS

Languages: TypeScript · JavaScript · SQL | **Frontend:** React · Next.js · Tailwind CSS · Redux / Zustand

Backend: Node.js · NestJS · Express · tRPC · REST APIs | **Databases:** PostgreSQL · TypeORM · Drizzle · MongoDB · NoSQL

Infra & Testing: Docker · AWS (ECS, RDS, S3, EC2) · Vercel · GitHub Actions · CI/CD · Jest | **Architecture:** Clean Architecture · Vertical Slice Architecture · SOLID · DDD

Web Accessibility: WCAG 2.1 (keyboard navigation, color contrast, semantic HTML, ARIA attributes)

AI-Assisted Development: AI: LLM integration · RAG · MCP · Structured outputs · Claude Code · Cursor · Spec-Driven Development (SDD) · Multi-agent workflows

EDUCATION

Associates In Web Development — National University of Comahue 2019 – 2022

Systems & Computer Engineering — Universidad de los Andes (ongoing) 2023 – Present